ReHQJ: the HexOpt Paper and conspire

autotwin/automesh issue #768

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Sources

This note compares the HexOpt objective in two sources.

- **Paper.** H. Tong and Y. J. Zhang, "Fast and Robust Hexahedral Mesh Optimization via Augmented Lagrangian, L-BFGS, and Line Search," arXiv:2410.11656, Equations (2.5), (2.6), (2.8), and (2.9).
- **conspire.** Branch `msj` at commit `515c78cee`, draft mrbuche/conspire.rs#251, file `src/geometry/mesh/quality/improve/msj/mod.rs`, functions `evaluate` and `terms`.

h is one hex element of the mesh H, and $\bar{e}$ is its average edge length. $SJ(h)$ is the minimum of the scaled Jacobian over nine points of h: its eight corners and its body center (paper, Section 2.1). $J(h)$ is the minimum of the Jacobian over the same nine points. conspire computes both minimums in `terms`, each over all nine points. Θ is the threshold. Both sources hold $\bar{e}$ fixed when they take the gradient.

Definitions

The paper calls the per-hex term $ReHQJ(h, \Theta)$. This note writes it as $\psi(h)$, with subscript p for the paper and c for conspire. Both sources use the same term for an inverted hex:

$$\psi_{\mathrm{p}}(h) = \psi_{\mathrm{c}}(h) = J(h)/\bar{e}, \qquad J(h) \leq 0.$$

For $J(h) > 0$, the paper defines

$$\psi_{\mathrm{p}}(h) = \begin{cases} \bar{e}^2\, SJ(h), & SJ(h) \leq \Theta, \\ \Theta, & SJ(h) > \Theta, \end{cases}$$

and conspire implements

$$\psi_{\mathrm{c}}(h) = \bar{e}^2 \min\big(SJ(h), \Theta\big).$$

The two differ only above Θ, where $\psi_{\mathrm{p}} = \Theta$ and $\psi_{\mathrm{c}} = \Theta\,\bar{e}^2$. ψ_{c} is one expression: $\bar{e}^2$ times SJ clamped at Θ. ψ_{p} has no such form unless $\bar{e} = 1$.

Units

$J = \det(e_0, e_1, e_2)$, where each edge vector has units of length. SJ and Θ have no units.

case	ψ_p	ψ_c
$J \le 0$	length^2	length^2
$J > 0$, $SJ \le \Theta$	length^2	length^2
$SJ > \Theta$	**none**	length^2

The minimized function

At each threshold Θ, conspire minimizes

$$L = -\sum_{h \in H} \psi_\mathrm{c}(h) + \sum_i \Big[\lambda_i\, r_i + \frac{\rho}{2}\, r_i^2\Big].$$

- The first sum has one term per hex, with ψ_c as defined above.
- The second sum is an augmented Lagrangian that holds the boundary on the target STL. $r_i = n_i \cdot x - d_i$ is the signed distance from a boundary node x to the plane of one STL triangle. Each boundary node gets up to four planes: the plane of its closest triangle, plus the planes of nearby triangles whose normals differ from those already kept by more than 30°.
- After each L-BFGS run, conspire updates each multiplier, $\lambda_i \leftarrow \lambda_i + \rho\, r_i$. ρ starts at 100 and doubles each time the mesh meets Θ with the boundary on the STL.
- r_i and λ_i have units of length, and ρ has none. Both constraint terms have units of length^2.

The paper's Equation (2.8) has the same structure. Its constraint is $Z_k - Z_k^t$, the offset from each boundary vertex to its closest point on the target surface. Its Equation (2.9) updates the multipliers the same way, and it doubles ρ each time the minimum scaled Jacobian reaches Θ. The paper's L uses ψ_p in place of ψ_c. The next section returns to that difference.

Behavior at the threshold

1. **Continuity.** At $SJ = \Theta$, both forms below Θ give $\Theta\,\bar{e}^2$. Above Θ, ψ_c gives the same value, so it is continuous at Θ. Above Θ, ψ_p gives Θ, so it jumps by $\Theta(\bar{e}^2 - 1)$ unless $\bar{e} = 1$. Neither form is differentiable at Θ. The slope drops to zero there in both, by design.

2. **The sign of the jump.** L, above, sums $-\psi$ over all hexes. Under ψ_p, a hex that rises above Θ changes L by $+\Theta(\bar{e}^2 - 1)$. For $\bar{e} > 1$, L rises. The Armijo condition requires L to fall, so the jump counts against every step that lifts such a hex above Θ. For $\bar{e} < 1$, L falls, so the jump rewards the crossing instead. The sign depends on the length unit of the mesh.

   - On the snapped bone octree in the reference frame, $\bar{e}$ runs from 0.644 to 2.81, with a median of 1.253. The same mesh has hexes on both sides of $\bar{e} = 1$.
   - On the scale-5 bone mesh in the STL's own units, $\bar{e}$ runs from 0.009 to 0.034.

   Under ψ_c, the crossing changes L by zero.

3. **Units.** ψ_c has units of length^2 in every case. conspire's types enforce this: each term is a `Quantity<Area>`.

For a hex that stays above Θ through a step, both forms are constants. They give the same gradient and the same energy differences. The forms differ only when a hex crosses Θ.